

## Full Length Article

## HAD-MSF: Multi-domain neural fusion with state-space and graph modeling for hyperspectral anomaly detection

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## ABSTRACT

Hyperspectral anomaly detection (HAD) remains challenging because spatial, spectral, and frequency dependencies coexist and exhibit heterogeneous characteristics. Existing CNN-, Transformer-, and GCN-based approaches often rely on a single modeling paradigm, which may limit their ability to fully exploit these complementary structures. In this work, we propose HAD-MSF, a neural reconstruction framework that integrates multiple domain-specific representations within a unified architecture. An Adaptive Wavelet Approximation Transform (AdaWAT) separates hyperspectral cubes into low- and high-frequency components, which are processed by a state-space-driven Spatial-Frequency Mamba module to capture long-range dependencies with linear complexity. In parallel, a Spatial-Spectral Graph Convolution branch models pixel topology together with inter-band correlations, enhancing spectral discrimination. A lightweight gating mechanism adaptively fuses the two representations into a compact reconstruction network, where anomalies are identified from residuals. Experiments on three remote sensing and two medical hyperspectral datasets show that HAD-MSF achieves consistently competitive detection performance with relatively low arithmetic complexity. Additional analyses on component design, threshold selection, robustness, cross-scene transfer, and full-band settings further characterize the effectiveness and practical properties of the proposed framework. These results demonstrate the effectiveness of integrating wavelet, state-space, and graph modeling for hyperspectral anomaly detection.

## 1. Introduction

Hyperspectral anomaly detection (HAD) has broad applications in remote sensing (Liu et al., 2025b; Xie et al., 2019), environmental monitoring (Goudemant et al., 2024; Zare-Baghbidi & Homayouni, 2013), military reconnaissance (Hupel & Stütz, 2022; Tiwari et al., 2011), and medical imaging (Wang et al., 2020). By capturing hundreds of contiguous spectral bands, hyperspectral imaging (HSI) provides rich spatial-spectral information for detecting subtle anomalies, such as small targets in cluttered backgrounds (Gross et al., 2021; Shimoni et al., 2019) and malignant tissues in medical images (Liu et al., 2026; Zhang et al., 2024a).

HAD nevertheless remains challenging because of high spectral dimensionality, heterogeneous backgrounds, and the coexistence of spatial, spectral, and frequency dependencies. Spatial dependencies characterize local and long-range relations among pixels; spectral dependencies describe inter-band correlations, including non-local and non-Euclidean relations; and frequency dependencies distinguish smooth

background structures from localized high-frequency variations associated with small targets, boundaries, and noise (Lian et al., 2024; Tu et al., 2024). These structurally different dependencies may not be adequately modeled by a single uniform operator.

Traditional methods were primarily based on statistical modeling and subspace analysis. The well-known Reed-Xiaoli (RX) detector (Reed & Yu, 1990), for example, assumes a multivariate Gaussian background and flags outliers as anomalies. Later approaches incorporated sparse and low-rank priors through matrix decomposition or tensor formulations to better preserve the intrinsic structure of hyperspectral cubes. While effective to some extent, these methods rely on handcrafted assumptions and linear models, limiting their ability to cope with nonlinear and heterogeneous data.

With the rise of deep learning, HAD has benefited from nonlinear representation learning and end-to-end background reconstruction. Convolutional neural networks (CNNs) (Cheng et al., 2024) capture local spatial patterns but are inherently restricted by limited receptive fields. Transformers extend receptive fields through self-attention and

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capture long-range dependencies (Lian et al., 2024), but their quadratic complexity hinders scalability to large hyperspectral data. Recently, state-space models such as Mamba (Liu et al., 2025a; Zhu et al., 2024) have emerged as efficient alternatives, offering global receptive fields with linear complexity. At the same time, graph-based methods have shown potential for modeling non-Euclidean spectral relations (Sheng et al., 2024; Tu et al., 2024). However, most prior works employ only a single modeling paradigm—convolution, attention, or graph learning—which may limit their ability to simultaneously exploit spatial, spectral, and frequency cues.

In particular, the spectral dimension introduces additional complexity: correlations may exist between distant wavelengths due to biochemical absorption or scattering effects (e.g., water, hemoglobin, chlorophyll). While graph-based models can capture such dependencies, they often overlook frequency cues and seldom integrate seamlessly with spatial modeling. Similarly, frequency-aware decomposition can enhance anomaly saliency, but without explicit spectral-topological modeling, important inter-band relationships may still be underexplored. This highlights the need for a unified neural framework that jointly exploits spatial, spectral, and frequency structures for effective anomaly detection.

The motivation of HAD-MSF is to address the heterogeneous spatial, spectral, and frequency dependencies of hyperspectral data using complementary inductive biases. In reconstruction-based HAD, background structures are typically spatially coherent and dominated by smooth components, whereas anomalies often appear as localized frequency perturbations and deviations from surrounding spectral relations. Accordingly, HAD-MSF combines frequency decomposition, long-range state-space modeling, and graph propagation within a unified reconstruction framework rather than relying on a single feature extractor.

To this end, we propose HAD-MSF, a neural reconstruction framework that integrates complementary domain-specific representations within a unified architecture. First, AdaWAT decomposes hyperspectral cubes into low-frequency structures and high-frequency details, providing anomaly-sensitive frequency separation. These components are further processed by a Spatial-Frequency Mamba module, which leverages state-space modeling to capture long-range dependencies with linear complexity. In parallel, a Spatial-Spectral Graph Convolution Network models pixel-level topology together with inter-band correlations. Finally, a lightweight gating mechanism adaptively fuses the outputs into a compact reconstruction network, where anomalies are identified as residuals.

HAD-MSF is trained without anomaly annotations. The soft anomaly weights used during separation training are generated online from reconstruction residuals rather than ground-truth labels, preserving the unsupervised setting of the proposed framework.

Experiments on three remote sensing and two medical hyperspectral datasets demonstrate that HAD-MSF provides competitive and well-balanced detection performance across multiple evaluation criteria, while the additional analyses verify the contributions of the proposed components and reveal the remaining challenge of cross-scene transfer.

The main contributions of this work are summarized as follows:

- We formulate HAD as a heterogeneous structure-modeling problem and propose HAD-MSF to jointly model spatial-frequency and spatial-spectral dependencies within an unsupervised reconstruction framework.
- We develop a spatial-frequency track that combines AdaWAT decomposition with SF-Mamba to model smooth background structures and localized anomaly-sensitive details while capturing long-range spatial context.
- We construct a complementary spatial-spectral graph track to model neighborhood topology and inter-band relations, and introduce adaptive gating to fuse the two branches during reconstruction.
- Extensive experiments on three remote sensing and two medical hyperspectral datasets, together with component, fusion, full-band, ro-

bustness, and cross-scene analyses, validate the effectiveness of the proposed framework across heterogeneous hyperspectral scenarios.

The remainder of this paper is organized as follows. Section 2 reviews related HAD methods. Section 3 presents the proposed HAD-MSF framework. Section 4 reports the experimental results and analysis. Section 5 concludes the paper and discusses limitations and future work.

## 2. Related work

### 2.1. Statistical and subspace approaches

Early hyperspectral anomaly detection (HAD) approaches were dominated by statistical modeling and subspace representations. The Reed-Xiao (RX) detector (Reed & Yu, 1990), for example, assumes a multivariate Gaussian distribution of the background and flags samples that deviate from this distribution as anomalies. To overcome its limitations, subsequent methods introduced sparse and low-rank priors via matrix decomposition or tensor formulations, aiming to preserve the intrinsic geometry of hyperspectral cubes (Hupel & Stütz, 2022; Tiwari et al., 2011; Xu et al., 2026a; Zare-Baghibidi & Homayouni, 2013). These approaches improved robustness under certain conditions, but their reliance on handcrafted assumptions and linear formulations limited their ability to model highly nonlinear or heterogeneous data distributions. Recent work has therefore combined classical priors with learned representations. Dynamic low-rank and sparse priors constrained deep autoencoders integrate interpretable low-rank/sparse structure with nonlinear reconstruction (Lin et al., 2024), while sparse-collaborative representation jointly exploits sparse coding and collaborative background reconstruction (Lin et al., 2023).

### 2.2. Deep neural architectures

The advent of deep learning has greatly advanced HAD by enabling nonlinear feature learning and end-to-end background reconstruction. Convolutional neural networks (CNNs) have been applied to capture local spatial structures and enhance detection accuracy (Cheng et al., 2024; Goudemant et al., 2024). Recent reconstruction-oriented HAD methods have explored increasingly structured background modeling strategies. GAN-based approaches include FTSGAN, which reformulates background reconstruction as a frequency-to-spectrum mapping problem (Wang et al., 2023a), and CL-CaGAN, which incorporates adversarial continual learning to improve cross-domain anomaly detection (Wang et al., 2024c). Blind-spot and blind-block reconstruction methods further reduce anomaly over-reconstruction by restricting direct access to central pixels or regions: PDBSNet combines pixel-shuffle downsampling with blind-spot reconstruction (Wang et al., 2023c), BockNet adopts blind-block reconstruction with contextual constraints (Wang et al., 2023b), DirectNet introduces a sliding dual-window-inspired reconstruction strategy (Wang et al., 2024b), and PUNNet exploits global feature-guided blind-spot reconstruction (Wang et al., 2024a). NL2Net further couples local masked/dilated convolution with non-local feature extraction for self-supervised background modeling (Wang et al., 2025). More recent methods emphasize explicit anomaly suppression during reconstruction through multi-scale memory modeling (Huo et al., 2026a), dual-stream background modeling (Huo et al., 2026c), and dual-domain background representation (Huo et al., 2026b). These studies indicate that effective HAD increasingly relies on structured background representation together with explicit mechanisms for limiting anomaly over-reconstruction.

However, CNNs are inherently constrained by local receptive fields, which restrict their capability to model global dependencies. Transformers alleviate this issue by employing self-attention to capture long-range correlations (Lian et al., 2024; Liu et al., 2025b), but their quadratic complexity makes them computationally expensive, especially for large-scale hyperspectral data. Recently, state-space models such as

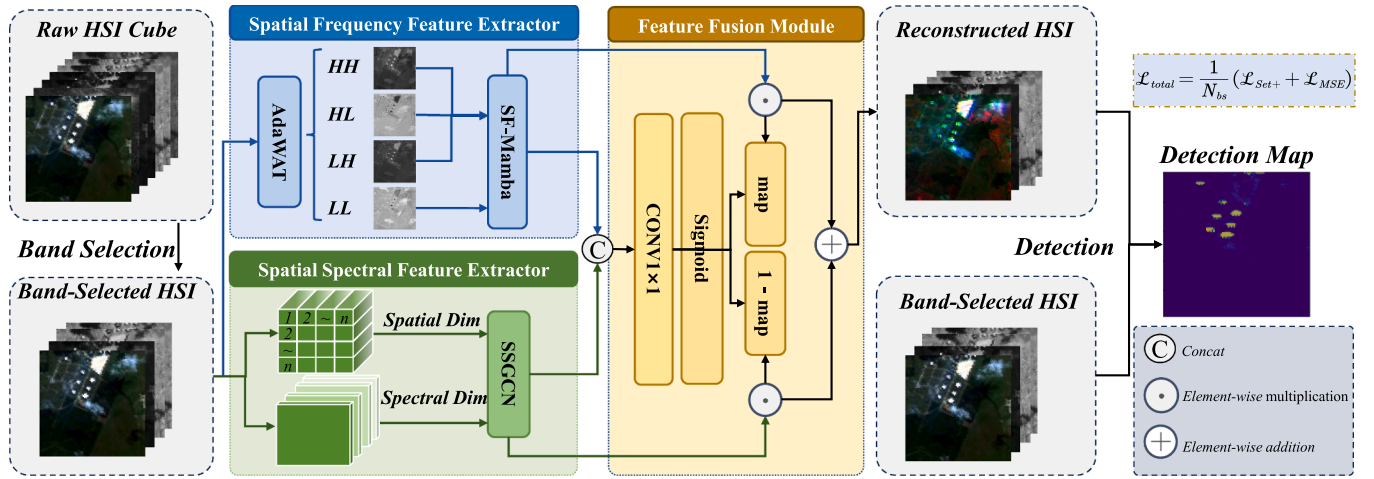


Fig. 1. Overall architecture of the proposed HAD-MSF framework. It consists of a spatial–frequency track and a spatial–spectral track, whose outputs are adaptively fused to generate the reconstruction for anomaly detection.

Mamba (Liu et al., 2025a; Zhu et al., 2024) have emerged as efficient alternatives, providing global receptive fields with linear complexity and showing strong potential for high-dimensional imaging tasks.

Beyond remote sensing, HAD is also attracting growing attention in medical hyperspectral imaging. Chen et al. (2026) proposed a self-supervised visual auto-regressive framework that integrates multi-scale modeling, graph-based spectral representation, and feature fusion for medical hyperspectral anomaly detection, further demonstrating the value of jointly exploiting spatial and spectral structures in heterogeneous imaging scenarios.

### 2.3. Graph-based spectral methods

The spectral dimension poses additional challenges because spectral bands exhibit structured and potentially non-local dependencies. Correlations between wavelengths may arise from material- or tissue-specific absorption and scattering characteristics, resulting in spectral relationships that cannot always be adequately represented by purely spatial modeling (Cui et al., 2022; Peyghambari & Zhang, 2021). To capture these dependencies, graph-based methods have been proposed, in which pixels or bands are represented as nodes and relationships are encoded as edges (Sheng et al., 2024; Tu et al., 2024). Graph convolutional networks (GCNs) enable modeling of inter-band correlations and non-local relations, thereby complementing spatial learning. However, these approaches typically do not incorporate frequency cues and are rarely integrated seamlessly with spatial modeling.

Recent graph-based HSI studies have further explored adaptive and label-free graph learning. AF2GNN introduces adaptive graph filters and aggregator fusion for hyperspectral image classification (Ding et al., 2022), while AHSGC (Ding et al., 2025a), SLGCG (Ding et al., 2025b), and DSGC (Liang et al., 2026) explore label-free HSI clustering through adaptive, self-supervised, or dynamic graph modeling.

Different from the clustering-oriented AHSGC, SLGCG, and DSGC, HAD-MSF performs unsupervised anomaly detection through background reconstruction and residual scoring. Its graph branch serves as a structural constraint complementary to the spatial–frequency track rather than generating semantic clusters or class pseudo-labels.

### 2.4. Multi-domain fusion strategies

Although significant progress has been made, many existing methods still rely primarily on a single modeling paradigm, such as convolution, attention, or graph learning, which may limit the joint exploitation of complementary spatial, spectral, and frequency structures (Lian et al., 2024; Tu et al., 2024; Xu et al., 2026b; Zhang

et al., 2024a). Recent studies have therefore explored heterogeneous and multi-domain representations. For example, D2G-Net combines disentangled spatial–spectral representations with state-space and graph modeling for cross-domain HAD (Xu et al., 2025), while DDBMNet and dual-stream background modeling exploit complementary representations to improve background reconstruction and anomaly suppression (Huo et al., 2026b,c). Nevertheless, the integration of explicit frequency decomposition, state-space modeling, and spectral graph learning within a unified reconstruction framework remains underexplored, motivating the proposed HAD-MSF.

Related reconstruction-based approaches have also explored background–anomaly separation through background purification, prototype-guided attribution, context-aware suppression, and memory-based modeling. Attribution-driven background modeling uses representative background prototypes and surrounding context to constrain anomalous reconstruction (Wang et al., 2026), whereas multi-scale memory modeling with separation training learns background-oriented representations while reducing anomaly contamination (Huo et al., 2026a). These strategies demonstrate the importance of explicitly controlling anomaly information during background reconstruction.

In contrast, HAD-MSF does not construct a purified background library or assign hard background/anomaly pseudo-labels. Instead, its soft mask is generated online from reconstruction residuals and serves as a continuous weighting mechanism for background reconstruction and anomaly suppression. Meanwhile, SSGCN models spatial and inter-band topology, whereas SF-Mamba captures long-range spatial dependencies and frequency-aware representations. HAD-MSF therefore combines residual-guided training with complementary spatial–spectral and spatial–frequency modeling rather than using the mask as an explicit background-category estimator.

## 3. The proposed HAD-MSF framework

### 3.1. Overview

A central challenge in hyperspectral anomaly detection is that spatial, spectral, and frequency dependencies exhibit different structural characteristics. Most existing approaches rely primarily on a single modeling paradigm, which may not fully exploit these complementary cues. To address this limitation, we propose HAD-MSF, a reconstruction-based framework that decouples spatial, spectral, and frequency representations and integrates them through adaptive fusion.

As illustrated in Fig. 1, the input hyperspectral cube  $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$  first undergoes band selection to reduce spectral redundancy and noise. The resulting cube  $\mathbf{X}_{bs}$  is processed by two complementary tracks: the

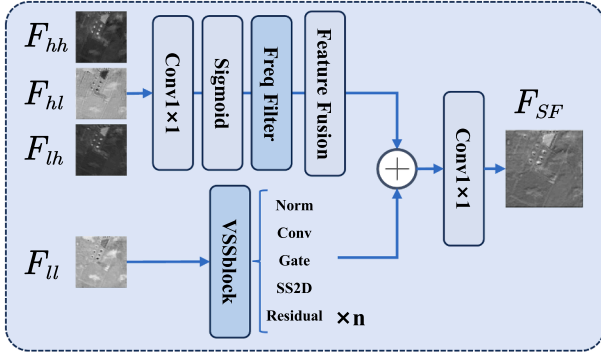


Fig. 2. Structure of the spatial-frequency Mamba (SF-Mamba) block. It integrates low-frequency global context (via VSS) and high-frequency details (via learnable gates).

*spatial-frequency track* captures frequency-aware and long-range dependencies, while the *spatial-spectral track* models pixel topology and inter-band correlations. Their outputs are adaptively fused to reconstruct  $\hat{\mathbf{X}}$ , from which anomaly scores are derived using reconstruction residuals.

The architecture can be interpreted through a structure-oriented decomposition of the observed hyperspectral cube,  $\mathbf{X}_{bs} = \mathbf{B} + \mathbf{A} + \mathbf{N}$ , where  $\mathbf{B}$ ,  $\mathbf{A}$ , and  $\mathbf{N}$  denote the dominant background, anomalous deviations, and noise, respectively. This expression is used only to motivate the architectural design rather than as a strict generative assumption. Background regions are generally more spatially coherent and spectrally consistent, whereas anomalies may introduce localized frequency responses and deviations from surrounding spectral relations.

Accordingly, AdaWAT and SF-Mamba jointly model smooth/background and localized frequency-aware structures with long-range spatial context, while SSGCN provides complementary spatial-neighborhood and inter-band graph constraints. Their outputs are combined through a data-dependent gating mechanism, allowing the reconstruction network to adaptively balance the two structural representations rather than relying on a fixed fusion rule. Therefore, HAD-MSF is designed as a complementary spatial-frequency/spatial-spectral decomposition for reconstruction-based HAD, whose effectiveness is further examined through the branch and fusion ablations in Section 4.

### 3.2. Band selection and preprocessing

Given an input hyperspectral image  $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ , we apply a unified band-selection preprocessing step before feature extraction. The purpose is twofold: 1) to suppress noisy or low-informative channels, including water-absorption bands when present; and 2) to reduce spectral redundancy and the computational burden of subsequent models. In this work, we adopt the SIFDR strategy following Liu et al. (2024), Zhang et al. (2024b) to select a compact subset of informative and distinctive bands from the original spectrum, yielding a reduced cube

$$\mathbf{X}_{bs} \in \mathbb{R}^{H \times W \times N_{bs}}, \quad N_{bs} = 64. \quad (1)$$

The selected cube  $\mathbf{X}_{bs}$  serves as the common input to all subsequent modules. Importantly, this band-selection step is applied identically to HAD-MSF and all compared methods, including classical, representation-based, and learning-based baselines. Therefore, the 64-band input is a common preprocessing setting rather than a model-specific optimization tailored to HAD-MSF. In essence, SIFDR ranks bands by jointly considering spatial-spectral information content and inter-band redundancy, and then retains a compact subset of informative and complementary channels (see Fig. 2).

### 3.3. Spatial-frequency track: AdaWAT and SF-Mamba

Hyperspectral images typically contain smooth low-frequency backgrounds together with discriminative high-frequency variations. Treating all frequencies uniformly may oversmooth anomalies or amplify noise. To address this, we introduce the Adaptive Wavelet Approximation Transform (AdaWAT):

$$(F_{LL}, F_{LH}, F_{HL}, F_{HH}) = \text{AdaWAT}(\mathbf{X}_{bs}), \quad (2)$$

where  $F_{LL}$  denotes approximation coefficients and  $\{F_{LH}, F_{HL}, F_{HH}\}$  represent horizontal, vertical, and diagonal details.

Unlike traditional Haar transforms with fixed kernels, AdaWAT employs *learnable depthwise convolutions* with  $3 \times 3$  kernels:

$$F_d(i, j) = \sum_{u=-1}^1 \sum_{v=-1}^1 \mathbf{K}_d(u, v) \cdot X_{bs}(i + u, j + v), \quad d \in \{LL, LH, HL, HH\}, \quad (3)$$

where  $\mathbf{K}_d$  is initialized with Haar-like filters and optimized during training. This design provides interpretable starting points that encode basic frequency structures, while enabling the network to adaptively refine them according to dataset-specific statistics. As a result, AdaWAT learns data-driven frequency decompositions rather than relying solely on handcrafted priors. In practice, visualizations of the learned kernels show clear deviations from fixed filters, supporting the claim that AdaWAT captures task-relevant frequency responses. Unlike generic adaptive filtering, AdaWAT retains an explicit four-subband wavelet-like decomposition ( $LL/LH/HL/HH$ ), initializes depthwise kernels with Haar-like priors, and jointly optimizes them for anomaly reconstruction. It therefore provides a task-oriented frequency decomposition rather than unrestricted feature filtering. This design also differs from frequency-to-spectrum approaches such as FTSGAN (Wang et al., 2023a), which use adversarial mapping between frequency and spectral domains.

Frequency decomposition is particularly relevant to HAD because background regions are often dominated by smooth low-frequency structures, whereas small anomalies, boundaries, and local spectral variations may produce stronger high-frequency responses. Explicit separation helps preserve anomaly-sensitive details that may otherwise be oversmoothed during background reconstruction. Since high-frequency components may also contain noise, adaptive high-frequency weighting and complementary SSGCN constraints are used to suppress irrelevant responses.

In the current implementation, AdaWAT uses single-level learnable depthwise filters with a fixed kernel size of  $3 \times 3$ . The filters are initialized with Haar-like priors and then jointly optimized with the network. Therefore, AdaWAT introduces only limited manual tuning requirements in practice, with initialization strategy and kernel size being the main configurable design choices.

The low-frequency branch  $F_{LL}$  is further processed by a Visual State-Space (VSS) module, which captures long-range dependencies with *linear complexity*. The module follows the standard state-space formulation:

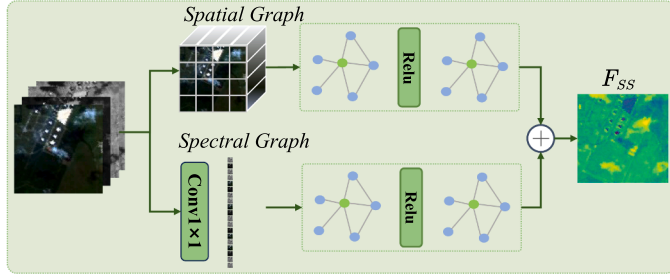
$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t + \mathbf{D}x_t, \quad (4)$$

where  $h_t$  is the hidden state,  $x_t$  the input, and  $y_t$  the output.  $\bar{\mathbf{A}}$  and  $\bar{\mathbf{B}}$  are input-dependent discretizations of continuous-time state matrices, while  $\mathbf{C}$  and  $\mathbf{D}$  are output projection matrices. Compared with Transformers that incur quadratic  $O(n^2)$  complexity, the VSS module captures global dependencies with linear  $O(n)$  complexity. The final SF-Mamba implementation uses a single VSS block, with deeper configurations further evaluated in the ablation study.

The high-frequency details are combined through a frequency-aware weighting mechanism:

$$F_{HF} = \frac{1}{3} \sum_{d \in \{LH, HL, HH\}} \sigma(\mathbf{W}_d) \odot F_d, \quad (5)$$





**Fig. 3.** Structure of the spatial-spectral graph convolution network (SSGCN). It jointly models pixel adjacency in the spatial domain and band correlations in the spectral domain.

where  $\sigma(\cdot)$  denotes the sigmoid activation and  $\mathbf{W}_d$  are learnable  $\text{Conv}_{1 \times 1}$  filters that generate adaptive weights. Although reminiscent of a gating operation, this formulation is essentially a weighted averaging of the three high-frequency sub-bands, providing a simple yet effective alternative to more complex channel-attention mechanisms (see Fig. 3).

Finally, the spatial-frequency representation is obtained as

$$F_{SF} = \text{Conv}_{1 \times 1}(\text{VSS}(F_{LL}) + F_{HF}). \quad (6)$$

### 3.4. Spatial-spectral track: joint graph learning

Spectral signatures are inherently non-Euclidean, and anomalies are often characterized by subtle inter-band dependencies. To jointly capture pixel adjacency and band correlations, we design a Spatial-Spectral Graph Convolution Network (SSGCN).

**Spatial GCN.** Each pixel is treated as a node and connected bidirectionally to its four spatial neighbors (up, down, left, and right), forming the spatial graph  $\mathcal{G}_s$ . Diagonal connections are not included. Graph convolution updates node features as

$$H^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}} \hat{A}_s \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right), \quad (7)$$

where  $\hat{A}_s = A_s + I$  is the adjacency matrix with self-loops,  $H^{(l)}$  is the feature matrix at layer  $l$ , and  $W^{(l)}$  are trainable weights. After  $L$  layers of propagation, the resulting representation  $F_{\text{spatial}} = H^{(L)}$  encodes local topology and neighborhood consistency.

**Spectral GCN.** Within each pixel, spectral bands are modeled as nodes of a band graph  $\mathcal{G}_c$ . We adopt a fixed bidirectional chain adjacency in which each spectral band is connected to its immediately adjacent bands, reflecting the local smoothness commonly observed in hyperspectral sampling. Unlike learnable attention mechanisms, this prior adjacency is not updated during training; instead, the GCN optimizes feature representations of bands conditioned on this fixed graph structure. The spectral graph convolution is expressed as

$$Z^{(l+1)} = \sigma\left(\hat{D}_c^{-\frac{1}{2}} \hat{A}_c \hat{D}_c^{-\frac{1}{2}} Z^{(l)} W_c^{(l)}\right), \quad (8)$$

where  $\hat{A}_c = A_c + I$  is the normalized adjacency of the band graph. Through multi-hop propagation, even with fixed connectivity, non-local spectral dependencies can be captured indirectly, while complementary long-range relations are modeled more explicitly by the SF-Mamba branch. After  $L$  layers, the band-level representation is denoted as  $F_{\text{spectral}} = Z^{(L)}$ .

**Integration.** The two graph-based representations are combined as

$$F_{SS} = F_{\text{spatial}} + F_{\text{spectral}}, \quad (9)$$

where  $F_{\text{spatial}}$  captures spatial adjacency in the image plane and  $F_{\text{spectral}}$  encodes intra-pixel spectral dependencies. This additive fusion preserves both spatial smoothness and spectral distinctiveness, forming a complementary representation that is later integrated with the spatial-frequency track.

### 3.5. Adaptive fusion and training objective

The two tracks produce complementary representations, which are adaptively balanced by a lightweight gating network:

$$G = \sigma(\text{Conv}_{1 \times 1}([F_{SF}; F_{SS}])), \quad (10)$$

where  $[\cdot; \cdot]$  denotes feature concatenation and  $\sigma(\cdot)$  is the sigmoid activation. The reconstructed hyperspectral cube is then obtained as

$$\hat{\mathbf{X}} = G \odot F_{SF} + (1 - G) \odot F_{SS}, \quad (11)$$

where  $\hat{\mathbf{X}}$  denotes the reconstruction of the band-selected input  $\mathbf{X}_{bs}$ .

The training objective jointly promotes reconstruction fidelity and anomaly suppression. The basic reconstruction loss is defined as

$$\mathcal{L}_{\text{MSE}} = \|\mathbf{X}_{bs} - \hat{\mathbf{X}}\|_2^2, \quad (12)$$

which enforces spectral fidelity between the band-selected input  $\mathbf{X}_{bs}$  and its reconstruction  $\hat{\mathbf{X}}$ . Throughout the reconstruction and loss definitions,  $\mathbf{X}$  denotes the original HSI before band selection, whereas  $\mathbf{X}_{bs}$  is the actual network input and reconstruction target.

To construct the separation loss in a consistent manner, we first define the pixel-wise reconstruction residual as

$$R_{i,j} = \|\hat{\mathbf{X}}(i, j, :) - \mathbf{X}_{bs}(i, j, :)\|_2^2. \quad (13)$$

To further suppress anomaly responses, we adopt the enhanced separation training loss  $\mathcal{L}_{\text{SeT+}}$  following Liu et al. (2024). Based on the residual map  $R$ , a soft anomaly mask is constructed using min-max normalization:

$$M_{i,j} = \frac{R_{i,j} - \min(R)}{\max(R) - \min(R) + \epsilon}, \quad (14)$$

where  $\epsilon$  is a small constant introduced for numerical stability. The complementary background mask is then defined as

$$\bar{\mathbf{M}} = 1 - \mathbf{M}. \quad (15)$$

Using  $\mathbf{M}$  and  $\bar{\mathbf{M}}$ , the background reconstruction term is defined as

$$\mathcal{L}_{BR} = \frac{1}{S(\bar{\mathbf{M}}) + \epsilon} \|\hat{\mathbf{X}} - \mathbf{X}_{bs}\|_F^2 \odot \bar{\mathbf{M}}, \quad (16)$$

where  $S(\cdot)$  denotes the summation over all spatial elements and  $\odot$  denotes element-wise multiplication.

The anomaly suppression term is defined as

$$\mathcal{L}_{AS} = \frac{1}{S(\mathbf{M}) + \epsilon} \|\text{LoG}(\hat{\mathbf{X}}) \odot \mathbf{M}\|_F^2, \quad (17)$$

where  $\epsilon = 10^{-6}$  is used throughout the separation loss to avoid numerical instability, and  $\text{LoG}(\hat{\mathbf{X}})$  denotes the Laplacian-of-Gaussian filtered reconstruction.

The softened separation loss is therefore written as

$$\mathcal{L}_{\text{SeT+}} = \mathcal{L}_{BR} + \lambda \mathcal{L}_{AS}, \quad (18)$$

where  $\lambda = 10^{-3}$  is fixed across all experiments. The effective weights of  $\mathcal{L}_{\text{MSE}}$  and  $\mathcal{L}_{BR}$  are both set to 1, whereas  $\mathcal{L}_{AS}$  is weighted by  $\lambda = 10^{-3}$ . This configuration keeps reconstruction fidelity and background reconstruction as the dominant objectives, while using anomaly suppression as an auxiliary regularization term.

Compared with the original separation training strategy based on a hard binary mask, the present formulation employs a continuous soft anomaly mask. This avoids introducing an additional hard threshold during training and allows the contribution of each pixel to the separation objective to vary continuously according to its reconstruction residual, which is more suitable for heterogeneous hyperspectral scenes.

Finally, the total training objective is

$$\mathcal{L}_{\text{total}} = \frac{1}{N_{bs}} (\mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{SeT+}}), \quad (19)$$

where  $N_{bs}$  denotes the number of selected spectral bands. The normalization by  $N_{bs}$  keeps the overall loss scale comparable under the adopted band-selected input setting.

The optimization of HAD-MSF is fully unsupervised. Neither  $\mathcal{L}_{\text{MSE}}$  nor  $\mathcal{L}_{\text{SeT+}}$  requires ground-truth anomaly annotations, and the soft mask  $\mathbf{M}$  is generated online from the current reconstruction residual.

### 3.6. Test-time anomaly scoring

At test time, all network parameters are fixed, and the same pixel-wise reconstruction residual defined above is directly used as the anomaly score. Specifically,

$$S_{i,j} = R_{i,j} = \|\hat{\mathbf{X}}(i, j, :) - \mathbf{X}_{bs}(i, j, :)\|_2^2, \quad (20)$$

where a larger value indicates a higher anomaly likelihood. No anomaly labels are required for score generation. When normalized anomaly scores are needed for threshold-based analysis, the residual map is linearly scaled to  $[0, 1]$  using min-max normalization before applying the selected decision threshold.

### 3.7. Complexity analysis

Compared with Transformer-based detectors, whose self-attention scales as  $O(n^2)$  with the number of pixels/bands  $n$ , the SF-Mamba branch maintains linear  $O(n)$  complexity while retaining global receptive fields. In addition, the unified band-selection preprocessing reduces the input dimensionality from  $C$  to  $N_{bs}$ , which further lowers the computational burden of both the proposed model and the compared baselines. This linear-complexity state-space formulation reduces the asymptotic cost of global dependency modeling relative to quadratic self-attention and supports strong detection performance with relatively low arithmetic complexity in the controlled experiments. It also improves scalability to larger hyperspectral scenes, where quadratic self-attention can incur substantially higher computational costs.

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#### Algorithm 1 HAD-MSF for hyperspectral anomaly detection.

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**Require:** Hyperspectral cube  $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$

**Ensure:** Anomaly score map  $\mathbf{R} \in \mathbb{R}^{H \times W}$

- 1: **Band Selection:** Select informative bands from  $\mathbf{X}$  to reduce redundancy and noise.
  - 2: **Frequency Decomposition:** Apply AdaWAT to decompose selected  $\mathbf{X}_{bs}$  into low-frequency  $F_{LL}$  and high-frequency  $\{F_{LH}, F_{HL}, F_{HH}\}$ .
  - 3: **Spatial-Frequency Track:** Enhance  $F_{LL}$  with VSS; refine high-frequency components with learnable gates; fuse both to obtain  $F_{SF}$ .
  - 4: **Spatial-Spectral Track:** Construct pixel adjacency and band graphs; apply spatial GCN and spectral GCN; fuse outputs to obtain  $F_{SS}$ .
  - 5: **Adaptive Fusion:** Learn gating map  $G$  to adaptively combine  $F_{SF}$  and  $F_{SS}$ , yielding reconstruction  $\hat{\mathbf{X}}$ .
  - 6: **Anomaly Scoring:** Compute pixel-wise reconstruction error  $R(i, j)$  as the anomaly score.
  - 7: **return**  $\mathbf{R}$
- 

To summarize, the overall workflow of HAD-MSF is outlined in Algorithm 1. The procedure begins with band selection to reduce spectral redundancy, followed by AdaWAT-based frequency decomposition. The spatial-frequency and spatial-spectral tracks then extract complementary representations, which are adaptively fused via a gating mechanism to reconstruct the hyperspectral cube. Finally, anomaly scores are derived as reconstruction errors. This step-by-step description provides a concise operational view of the proposed framework, complementing the module-level details described above.

## 4. Experimental results and discussion

In this section, we conduct extensive experiments to comprehensively evaluate the proposed HAD-MSF framework. The goals are three-fold: (i) to demonstrate its competitive anomaly detection performance on both medical and remote sensing hyperspectral datasets; (ii) to examine its transferability across heterogeneous scenarios; and (iii) to analyze the contributions of each architectural component through ablation and robustness studies.

### 4.1. Experimental setup

**Datasets.** We evaluate HAD-MSF on three benchmark remote sensing datasets and two medical hyperspectral datasets. The two medical datasets, denoted as MHSI-1 and MHSI-2, are derived from the intraoperative brain hyperspectral imaging studies reported in Fabelo et al. (2019), Leon et al. (2023). For these medical datasets, the original spectra were preprocessed using the unsupervised spatial-spectral information fusion-based downsampled region (SIFDR) method (Zhang et al., 2024b) and reduced to 64 informative bands before evaluation. Together, these datasets cover heterogeneous imaging conditions and anomaly characteristics, enabling a comprehensive evaluation of the proposed method across both remote sensing and medical hyperspectral scenarios. The datasets are described as follows:

- **Pavia:** This dataset was acquired over Pavia, Italy, using the ROSIS sensor. It comprises 102 spectral bands with a spatial resolution of 1.3 m and an image size of  $100 \times 100$  pixels. Anomalies correspond to vehicles located on bridges and piers, with 43 anomalous pixels (0.43% of the image).
- **Coast:** Acquired by the AVIRIS sensor over the Texas coastal region, this dataset contains 204 spectral bands, a spatial resolution of 17.2 m, and an image size of  $100 \times 100$  pixels. The anomalies are coastal buildings, with 155 pixels labeled as anomalies (1.55% of the total).
- **Gulfport:** This dataset was collected by the AVIRIS sensor over Gulfport, Mississippi. It contains 191 spectral bands with a spatial resolution of 3.5 m and an image size of  $100 \times 100$  pixels. Anomalies are small airplanes distributed across the airport area, with 58 anomalous pixels (0.58%).
- **MHSI-1:** This medical dataset was collected from intraoperative hyperspectral imaging of human brain tissue using a visible-near-infrared camera. The image cube contains 64 spectral bands with a spectral range of 450–900 nm. The anomalies correspond to tumor regions, with 764 annotated anomalous pixels (7.64%).
- **MHSI-2:** This dataset is another intraoperative brain imaging scene acquired with the same system as MHSI-1. It contains 64 spectral bands covering 450–900 nm. Tumor tissue is annotated as anomalies, with 454 anomalous pixels (4.54%), providing a more challenging scenario with lower anomaly prevalence.

**Comparison methods.** We benchmark HAD-MSF against a diverse set of baselines covering three major paradigms. The statistical RX detector (Reed & Yu, 1990) serves as a classical reference. Representation-based methods including CRD (Li & Du, 2014), GTVLRR (Cheng & Wang, 2019), RPCA-RX (Candès et al., 2011), PTA (Li et al., 2020), LSMAD (Zhang et al., 2015), and LRASR (Xu et al., 2015) reconstruct the background under low-rank or sparse priors. Deep learning-based approaches such as Auto-AD (Wang et al., 2021), GAED (Xiang et al., 2022), GT-HAD (Lian et al., 2024), and BockNet (Wang et al., 2023b) leverage convolutional or attention-driven architectures to capture spatial-spectral dependencies, while PUNNet (Wang et al., 2024a) and NL2Net (Wang et al., 2025) further exploit blind-spot reconstruction and local-non-local contextual modeling, respectively.

**Implementation details.** The proposed HAD-MSF framework is implemented in PyTorch. Before training and evaluation, a unified band selec-

Table 1

Performance comparison of 14 anomaly detection methods on three remote sensing and two medical hyperspectral datasets. Eight evaluation metrics are reported for each dataset. The best results are in **bold**, and the second best are underlined.

| Dataset  | AUC values    | RX     | PTA           | GT-VLRR       | CRD    | RPCA-RX | LSMAD   | LRASR         | GAED    | GT-HAD          | Auto-AD | BockNet         | PUNNet        | NL2Net        | HAD-MSF          |
|----------|---------------|--------|---------------|---------------|--------|---------|---------|---------------|---------|-----------------|---------|-----------------|---------------|---------------|------------------|
| Pavia    | $(P_d, P_f)$  | 0.9934 | 0.9698        | 0.9950        | 0.9915 | 0.9918  | 0.9907  | 0.8860        | 0.9948  | <b>0.9992</b>   | 0.9900  | 0.9988          | 0.9982        | 0.9918        | 0.9991           |
|          | $(P_f, \tau)$ | 0.0258 | 0.0452        | 0.0134        | 0.0650 | 0.0145  | 0.0142  | 0.0580        | 0.0056  | 0.0022          | 0.0016  | 0.0004          | 0.0007        | 0.0006        | <b>0.0001</b>    |
|          | $(P_d, \tau)$ | 0.1601 | <b>0.3048</b> | 0.2536        | 0.2756 | 0.1702  | 0.1907  | 0.2107        | 0.2288  | 0.2381          | 0.0915  | 0.1449          | 0.1172        | 0.0881        | 0.0814           |
|          | TD            | 1.1534 | <b>1.2746</b> | 1.2485        | 1.2671 | 1.1620  | 1.1815  | 1.0967        | 1.2235  | 1.2373          | 1.0815  | 1.1438          | 1.1154        | 1.0798        | 1.0805           |
|          | BS            | 0.9676 | 0.9246        | 0.9816        | 0.9214 | 0.9773  | 0.9766  | 0.8280        | 0.9892  | 0.9970          | 0.9885  | 0.9984          | 0.9975        | 0.9912        | <b>0.9990</b>    |
|          | TD-BS         | 0.1343 | <b>0.2596</b> | 0.2402        | 0.2056 | 0.1794  | 0.1765  | 0.1527        | 0.2231  | 0.2359          | 0.0899  | 0.1445          | 0.1165        | 0.0875        | 0.0813           |
|          | ODP           | 1.1277 | 1.2294        | <b>1.2352</b> | 1.1971 | 1.1475  | 1.1673  | 1.0387        | 1.2179  | <u>1.2351</u>   | 1.0799  | 1.1434          | 1.1147        | 1.0793        | 1.0804           |
|          | SNPR          | 6.2072 | 6.7450        | 18.9493       | 3.9360 | 11.7536 | 13.4647 | 3.6330        | 40.7203 | <u>108.1964</u> | 58.4294 | <u>365.7545</u> | 171.0668      | 158.2001      | <b>1060.2634</b> |
|          | $(P_d, P_f)$  | 0.9906 | 0.9055        | 0.8510        | 0.9699 | 0.9922  | 0.9827  | 0.9863        | 0.9429  | 0.9968          | 0.9914  | 0.9917          | 0.9863        | 0.9798        | <b>0.9974</b>    |
|          | $(P_f, \tau)$ | 0.0571 | 0.2639        | 0.1890        | 0.0615 | 0.0367  | 0.0562  | 0.1156        | 0.0624  | <u>0.0094</u>   | 0.0049  | 0.0096          | 0.0040        | 0.0031        | <b>0.0002</b>    |
| Coast    | $(P_d, \tau)$ | 0.3177 | <b>0.5221</b> | 0.3839        | 0.2314 | 0.2705  | 0.3929  | 0.4956        | 0.3631  | 0.3747          | 0.3200  | 0.3665          | 0.2408        | 0.1545        | 0.2050           |
|          | TD            | 1.3083 | 1.4276        | 1.2349        | 1.2012 | 1.2627  | 1.3756  | <b>1.4819</b> | 1.3060  | 1.3716          | 1.3114  | 1.3582          | 1.2271        | 1.1342        | 1.2024           |
|          | BS            | 0.9335 | 0.6416        | 0.6620        | 0.9083 | 0.9555  | 0.9265  | 0.8707        | 0.8805  | 0.9874          | 0.9865  | 0.9822          | 0.9822        | 0.9767        | <b>0.9972</b>    |
|          | TD-BS         | 0.2607 | 0.2582        | 0.1949        | 0.1698 | 0.2338  | 0.3367  | <b>0.3799</b> | 0.3007  | <u>0.3653</u>   | 0.3151  | 0.3569          | 0.2368        | 0.1514        | 0.2048           |
|          | ODP           | 1.2512 | 1.1637        | 1.0459        | 1.1397 | 1.2260  | 1.3194  | <b>1.3663</b> | 1.2436  | <u>1.3622</u>   | 1.3065  | 1.3487          | 1.2230        | 1.1312        | 1.2022           |
|          | SNPR          | 5.5677 | 1.9783        | 2.0315        | 3.7604 | 7.3742  | 6.9864  | 4.2864        | 5.8161  | 39.9179         | 65.5858 | 38.2813         | 59.4703       | 50.0980       | <b>943.5954</b>  |
|          | $(P_d, P_f)$  | 0.9520 | 0.9955        | 0.9805        | 0.9814 | 0.9627  | 0.9866  | 0.9804        | 0.9680  | 0.9931          | 0.9738  | 0.9933          | 0.9599        | 0.9877        | <b>0.9984</b>    |
|          | $(P_f, \tau)$ | 0.0245 | 0.1881        | 0.0879        | 0.0510 | 0.0165  | 0.0539  | 0.1226        | 0.0348  | 0.0248          | 0.0258  | 0.0209          | 0.0085        | 0.0073        | <b>0.0002</b>    |
|          | $(P_d, \tau)$ | 0.0731 | <b>0.6900</b> | 0.3740        | 0.2830 | 0.0660  | 0.3628  | 0.4450        | 0.2932  | 0.2846          | 0.3131  | 0.4155          | 0.0876        | 0.1650        | 0.1460           |
|          | TD            | 1.0251 | <b>1.6855</b> | 1.3544        | 1.2644 | 1.0288  | 1.3494  | <u>1.4254</u> | 1.2612  | 1.2777          | 1.2869  | 1.4088          | 1.0475        | 1.1527        | 1.1444           |
| Gulfport | BS            | 0.9275 | 0.8074        | 0.8926        | 0.9304 | 0.9463  | 0.9327  | 0.8578        | 0.9333  | 0.9683          | 0.9480  | 0.9724          | 0.9514        | 0.9804        | <b>0.9982</b>    |
|          | TD-BS         | 0.0487 | <b>0.5018</b> | 0.2861        | 0.2320 | 0.0496  | 0.3089  | 0.3224        | 0.2584  | 0.2598          | 0.2873  | 0.3947          | 0.0791        | 0.1577        | 0.1458           |
|          | ODP           | 1.0006 | <b>1.4974</b> | 1.2666        | 1.2134 | 1.0123  | 1.2956  | 1.3028        | 1.2264  | 1.2529          | 1.2611  | 1.3880          | 1.0390        | 1.1454        | 1.1442           |
|          | SNPR          | 2.9884 | 3.6675        | 4.2564        | 5.5520 | 4.0133  | 6.7327  | 3.6291        | 8.4287  | 11.4710         | 12.1372 | 19.9005         | 10.3255       | 22.5853       | <b>679.6216</b>  |
|          | $(P_d, P_f)$  | 0.9555 | 0.8466        | 0.7078        | 0.9518 | 0.9485  | 0.9222  | 0.6979        | 0.8125  | 0.9270          | 0.6412  | 0.6323          | 0.8401        | 0.9707        | <b>0.9928</b>    |
|          | $(P_f, \tau)$ | 0.1771 | 0.3147        | 0.1044        | 0.1818 | 0.1420  | 0.1572  | 0.2172        | 0.1006  | 0.0406          | 0.0327  | 0.0564          | 0.0217        | 0.1606        | <b>0.0177</b>    |
|          | $(P_d, \tau)$ | 0.4411 | <b>0.5607</b> | 0.1451        | 0.3612 | 0.3670  | 0.3357  | 0.2655        | 0.2622  | 0.1688          | 0.0393  | 0.0694          | 0.0418        | 0.5396        | 0.3787           |
|          | TD            | 1.3966 | 1.4074        | 0.8529        | 1.3131 | 1.3155  | 1.2579  | 0.9634        | 1.0747  | 1.0958          | 0.6805  | 0.7016          | 0.8819        | 1.5103        | 1.3715           |
|          | BS            | 0.7783 | 0.5319        | 0.6034        | 0.7700 | 0.8065  | 0.7650  | 0.4807        | 0.7119  | 0.8864          | 0.6085  | 0.5759          | 0.8185        | 0.8101        | <b>0.9751</b>    |
|          | TD-BS         | 0.2640 | 0.2460        | 0.0407        | 0.1794 | 0.2250  | 0.1785  | 0.0483        | 0.1616  | 0.1282          | 0.0066  | 0.0130          | 0.0201        | 0.3789        | 0.3610           |
| MHSI-1   | ODP           | 1.2195 | 1.0927        | 0.7485        | 1.1312 | 1.1735  | 1.1007  | 0.7462        | 0.9741  | 1.0552          | 0.6478  | 0.6452          | 0.8602        | 1.3497        | <b>1.3538</b>    |
|          | SNPR          | 2.4907 | 1.7818        | 1.3905        | 1.9866 | 2.5849  | 2.1359  | 1.2222        | 2.6057  | <u>4.1600</u>   | 1.2010  | 1.2302          | 1.9279        | 3.3593        | <b>21.4260</b>   |
|          | $(P_d, P_f)$  | 0.8840 | 0.7464        | 0.5963        | 0.8830 | 0.8691  | 0.7343  | 0.5449        | 0.5524  | 0.6623          | 0.5763  | 0.5096          | 0.7301        | <b>0.9892</b> | 0.9749           |
|          | $(P_f, \tau)$ | 0.2132 | 0.3601        | 0.1497        | 0.1933 | 0.1511  | 0.1476  | 0.3069        | 0.1185  | 0.0862          | 0.0349  | 0.0254          | <b>0.0197</b> | 0.3961        | 0.1358           |
|          | $(P_d, \tau)$ | 0.3631 | 0.4862        | 0.1683        | 0.3074 | 0.2584  | 0.1976  | 0.3127        | 0.1116  | 0.1004          | 0.0361  | 0.0198          | 0.0262        | <b>0.7958</b> | 0.5974           |
|          | TD            | 1.2472 | 1.2325        | 0.7645        | 1.1904 | 1.1275  | 0.9319  | 0.8576        | 0.6640  | 0.7626          | 0.6124  | 0.5294          | 0.7562        | <b>1.7850</b> | <u>1.5724</u>    |
|          | BS            | 0.6708 | 0.3862        | 0.4466        | 0.6897 | 0.7180  | 0.5867  | 0.2380        | 0.4339  | 0.5761          | 0.5414  | 0.4842          | 0.7104        | 0.5932        | <b>0.8392</b>    |
|          | TD-BS         | 0.1499 | 0.1260        | 0.0185        | 0.1142 | 0.1073  | 0.0500  | 0.0058        | -0.0069 | 0.0142          | 0.0011  | -0.0055         | 0.0065        | <u>0.3997</u> | <b>0.4617</b>    |
|          | ODP           | 1.0339 | 0.8724        | 0.6148        | 0.9972 | 0.9764  | 0.7843  | 0.5507        | 0.5455  | 0.6765          | 0.5775  | 0.5041          | 0.7366        | 1.3890        | <b>1.4366</b>    |
|          | SNPR          | 1.7030 | 1.3500        | 1.1239        | 1.5908 | 1.7104  | 1.3388  | 1.0190        | 0.9421  | 1.1648          | 1.0327  | 0.7818          | 1.3306        | <u>2.0093</u> | <b>4.4003</b>    |

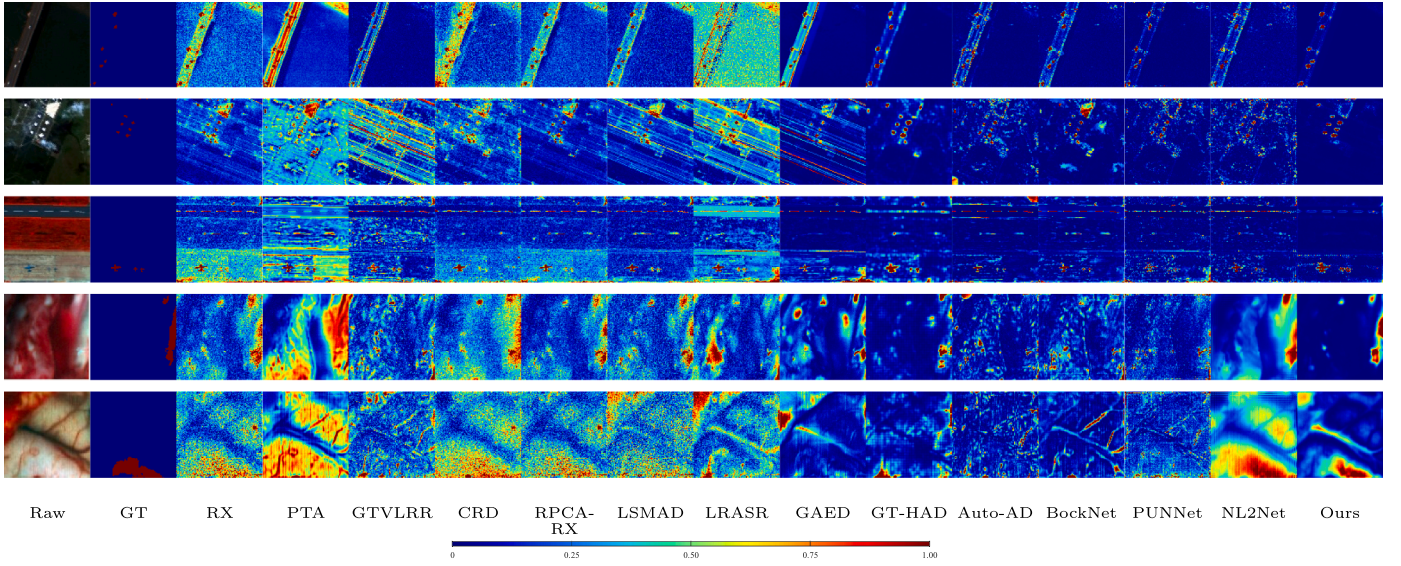


Fig. 4. Qualitative detection results on five hyperspectral datasets. Each row corresponds to one dataset (From top to bottom: Pavia, Coast, Gulfport, MHSI-1, MHSI-2), while each column displays the raw image, ground truth (GT), and results produced by 13 competing methods plus the proposed HAD-MSF.

tion procedure (Zhang et al., 2024b) is applied to all compared methods, reducing each dataset to  $N_{bs} = 64$  informative bands. This common pre-processing protocol controls the input dimensionality across methods and reduces potential confounding caused by different spectral input sizes. Our HAD-MSF is trained for 30 epochs using the Adam optimizer with a learning rate of  $1 \times 10^{-3}$  on an NVIDIA A100 GPU. Each hyperspectral image is processed as a full-image sample with a batch size of 1, and the random seed is fixed at 42 for reproducibility. The spatial graph adopts bidirectional four-neighborhood connectivity, while the spectral graph connects immediately adjacent bands in a bidirectional chain. The trade-off coefficient in  $\mathcal{L}_{\text{SeT}+}$  is fixed at  $\lambda = 10^{-3}$  across all experiments, and  $\epsilon = 10^{-6}$  is used to prevent division by zero in the separation loss. The same implementation configuration is used across the five datasets. To facilitate reproducibility, the implementation code will be released upon publication at <https://github.com/JzXu123/HAD-MSF>.

**Evaluation metrics.** We adopt both qualitative and quantitative evaluation protocols to comprehensively assess detection performance. Qualitative analysis includes *color anomaly maps*, which visualize background suppression and anomaly localization, and *boxplots*, which examine the separability between anomaly and background scores. Quantitative evaluation is based on the Area Under the ROC Curve (AUC). Specifically, three AUC measures are considered:  $AUC(P_d, P_f)$  (detection probability  $P_d$  versus false alarm rate  $P_f$ ),  $AUC(P_d, \tau)$  ( $P_d$  versus detection threshold  $\tau$ ), and  $AUC(P_f, \tau)$  ( $P_f$  versus  $\tau$ ). From these, five additional composite metrics are derived to capture complementary properties (see Fig. 5):

$$\begin{aligned}
 0 &\leq AUC_{\text{TD}} = AUC(P_d, P_f) + AUC(P_d, \tau) \leq 2, \\
 -1 &\leq AUC_{\text{BS}} = AUC(P_d, P_f) - AUC(P_f, \tau) \leq 1, \\
 -1 &\leq AUC_{\text{ODP}} = AUC(P_d, P_f) + AUC(P_d, \tau) - AUC(P_f, \tau) \leq 2, \\
 -1 &\leq AUC_{\text{TD-BS}} = AUC(P_d, \tau) - AUC(P_f, \tau) \leq 1, \\
 0 &\leq AUC_{\text{SNPR}} = \frac{AUC(P_d, \tau)}{AUC(P_f, \tau)}.
 \end{aligned} \tag{21}$$

Here,  $AUC_{\text{TD}}$  measures overall detection capacity,  $AUC_{\text{BS}}$  emphasizes background suppression,  $AUC_{\text{ODP}}$  and  $AUC_{\text{TD-BS}}$  capture trade-offs between detection and suppression, and  $AUC_{\text{SNPR}}$  quantifies signal-to-noise discrimination power. Lower  $AUC(P_f, \tau)$  and higher values of the other metrics indicate superior performance.

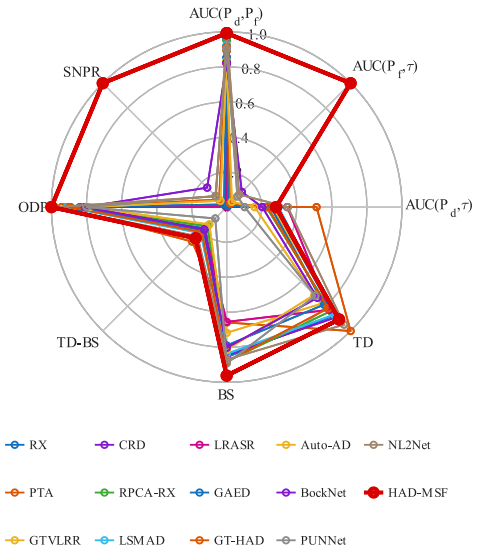


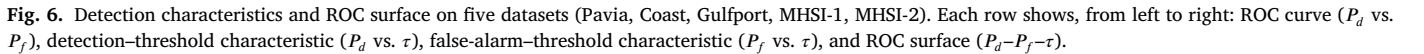
Fig. 5. Radar chart of 14 algorithms averaged over five hyperspectral datasets. Eight evaluation metrics are plotted after direction alignment and normalization. Concentric rings represent performance levels (0-1). Higher values indicate better performance.

#### 4.2. Quantitative evaluation and qualitative comparison

Table 1 compares fourteen anomaly detection methods across five hyperspectral datasets using eight complementary metrics. On the primary  $AUC(P_d, P_f)$  metric, HAD-MSF ranks first on Coast (0.9974), Gulfport (0.9984), and MHSI-1 (0.9928), and second on Pavia (0.9991) and MHSI-2 (0.9749). NL2Net achieves the highest  $AUC(P_d, P_f)$  on MHSI-2 (0.9892), whereas HAD-MSF obtains the best BS, TD-BS, ODP, and SNPR values on this dataset. Across the five datasets, HAD-MSF also attains the best  $AUC(P_f, \tau)$  on Pavia, Coast, and Gulfport and the best SNPR on all five datasets. Overall, these results indicate consistently strong and balanced performance across multiple evaluation criteria.

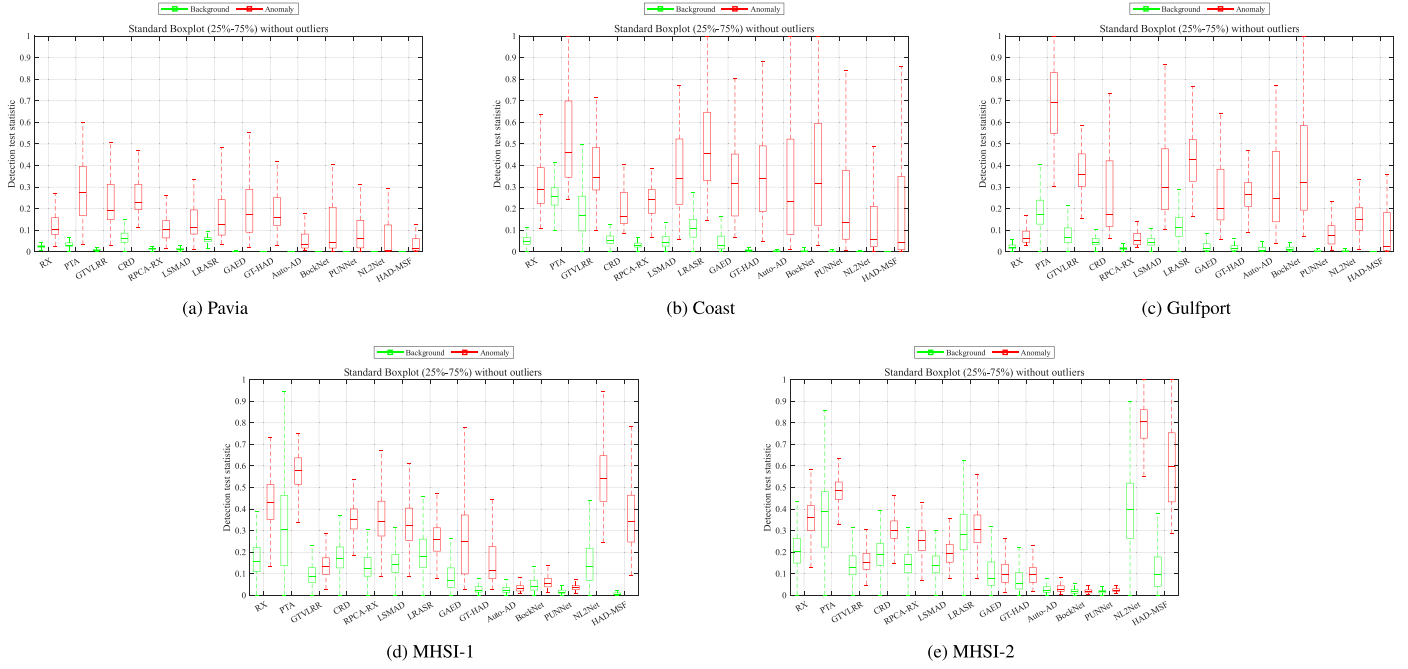
For fair comparison across heterogeneous metrics, each index was normalized to the range [0,1]. Specifically, for  $AUC(P_f, \tau)$ , where smaller values indicate better performance, we first applied an inversion ( $1/v$ ) to





and fusing spatial, spectral, and frequency structures yields a favorable balance across detection accuracy, background suppression, and anomaly separability.

In addition to the quantitative improvements, the qualitative detection maps in Fig. 4 further highlight the advantages of the proposed HAD-MSF. Traditional statistical and representation-based methods (e.g., RX, CRD, LRASR) often suffer from strong background clutter, leading to numerous false positives, while deep learning baselines such as Auto-AD and GT-HAD provide partial improvements but frequently miss subtle or small-scale anomalies. In contrast, HAD-MSF consistently produces anomaly maps with clean backgrounds and well-delineated target regions across both remote sensing and medical datasets. This performance gain can be attributed to its multi-domain design: the SF-Mamba branch effectively suppresses redundant low-frequency structures while preserving anomaly-related high-frequency cues, and the SSGCN branch captures non-local spectral correlations that



**Fig. 7.** Boxplots of anomaly score distributions for anomalous (red) and background (green) pixels across five datasets.

enhance anomaly separability. The complementary branches contribute to clearer anomaly–background separation across heterogeneous scenarios.

Beyond quantitative metrics, the AUC-based curves in Fig. 6 provide further evidence of the effectiveness of the proposed HAD-MSF. On the standard  $P_d$ – $P_f$  ROC curves, HAD-MSF is among the strongest methods across the five datasets. NL2Net is particularly competitive on MHSI-2, consistent with its higher  $AUC(P_d, P_f)$  on that scene, whereas HAD-MSF retains stronger background-suppression and composite separability metrics.

The  $P_d$ – $\tau$  and  $P_f$ – $\tau$  curves further illustrate the threshold-dependent detection and false-alarm trade-offs of HAD-MSF. The 3D ROC surfaces further illustrate method-dependent trade-offs under joint variations of  $P_f$  and  $\tau$ , with HAD-MSF maintaining a competitive operating region across heterogeneous scenes. This comprehensive advantage highlights that by decoupling and fusing spatial, spectral, and frequency structures, HAD-MSF not only improves raw detection accuracy but also maintains stable anomaly identification under diverse operating conditions, a property particularly important for both remote sensing and medical HSI applications.

In addition to ROC-based evaluations, the separability of anomaly and background scores was analyzed through boxplots (Fig. 7). Across all five datasets, HAD-MSF demonstrates consistently clear separation between anomalous and background score distributions. Specifically, the anomaly-score distributions are shifted toward higher values, while the background scores remain relatively compact and close to zero. This behavior indicates effective anomaly–background discrimination across heterogeneous scenes.

#### 4.3. Core ablation study

Table 2 presents the core ablation results averaged over five hyperspectral datasets. The full HAD-MSF achieves the strongest overall performance, attaining the highest values in  $AUC(P_d, P_f)$ ,  $AUC(P_d, \tau)$ ,  $AUC_{TD}$ ,  $AUC_{TD-BS}$ ,  $AUC_{ODP}$ , and  $AUC_{SNPR}$ . Removing AdaWAT, SF-Mamba, or SSGCN decreases the average  $AUC(P_d, P_f)$  from 0.9925 to 0.9843, 0.9865, and 0.9806, respectively, confirming the contributions of adaptive frequency decomposition, long-range spatial–frequency modeling, and spatial–spectral graph learning.

The adaptive fusion mechanism is also important. In the “Without adaptive fusion” variant, the two branch outputs are concatenated and projected by a  $1 \times 1$  convolution instead of being combined by the learned gating map, while all other components remain unchanged. This replacement reduces  $AUC(P_d, P_f)$  to 0.9703 and also degrades  $AUC_{TD-BS}$  and  $AUC_{ODP}$ . Moreover, either branch alone performs worse than the full model, indicating that the spatial–frequency and spatial–spectral tracks capture complementary anomaly cues that cannot be fully exploited by a single branch or simple feature combination.

The controlled ablations further show that the complete architecture benefits from the complementary roles of its components. The full model achieves an average  $AUC(P_d, P_f)$  of 0.9925, compared with 0.9796 and 0.9816 for the spatial–frequency and spatial–spectral tracks alone, respectively. Replacing adaptive gating with concatenation followed by a  $1 \times 1$  convolution further reduces the AUC to 0.9703, supporting the joint use of frequency decomposition, long-range state-space modeling, spatial–spectral topology, and adaptive fusion.

Overall, the ablation results support the complementary roles of AdaWAT, SF-Mamba, SSGCN, and adaptive fusion in the proposed multi-domain framework. We next examine the necessity of several component-level design choices of HAD-MSF.

#### 4.4. Component-level design comparison

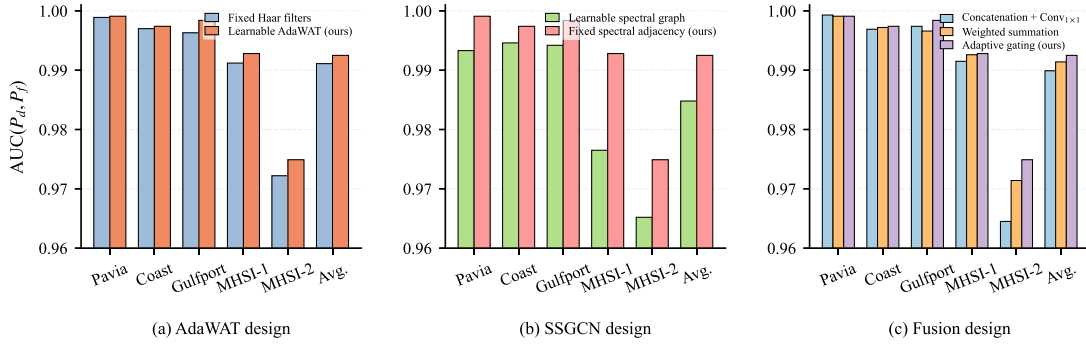
To further examine the necessity of several component-level design choices, Fig. 8 compares alternative implementations for AdaWAT, SSGCN, and the fusion module. In each comparison, only the target component is modified, while the remaining modules and training settings are kept identical to those of the full HAD-MSF model.

As shown in Fig. 8(a), replacing the learnable decomposition kernels with fixed Haar filters leads to a consistent performance drop on all five datasets, confirming the benefit of adaptive frequency decomposition over handcrafted wavelet priors. Fig. 8(b) shows that the adopted fixed spectral adjacency outperforms a learnable spectral graph, suggesting that a stable local spectral prior is more suitable than a fully learnable graph in the present setting. Fig. 8(c) further shows that adaptive gating achieves the best average performance among the compared fusion strategies, indicating that simple concatenation or weighted summation cannot fully exploit the complementarity between the two branches.

**Table 2**

Core ablation results of HAD-MSF averaged over five hyperspectral datasets. The full model combines AdaWAT, SF-Mamba, SSGCN, and adaptive gating. The best results are shown in **bold**, and the second-best results are underlined.

| Variant                           | Architecture setting    |                        |                             | Average performance across five datasets |                             |                           |                     |                     |                        |                      |                       |
|-----------------------------------|-------------------------|------------------------|-----------------------------|------------------------------------------|-----------------------------|---------------------------|---------------------|---------------------|------------------------|----------------------|-----------------------|
|                                   | Spatial–frequency track | Spatial–spectral track | Fusion                      | $AUC(P_d, P_f) \uparrow$                 | $AUC(P_f, \tau) \downarrow$ | $AUC(P_d, \tau) \uparrow$ | $AUC_{TD} \uparrow$ | $AUC_{RS} \uparrow$ | $AUC_{TD-RS} \uparrow$ | $AUC_{ODP} \uparrow$ | $AUC_{SNPR} \uparrow$ |
| Full HAD-MSF                      | AdaWAT + SF-Mamba       | SSGCN                  | Adaptive gating             | <b>0.9925</b>                            | 0.0308                      | <b>0.2817</b>             | <b>1.2742</b>       | 0.9617              | <b>0.2509</b>          | <b>1.2434</b>        | <b>541.8613</b>       |
| Without AdaWAT                    | Conv stem + SF-Mamba    | SSGCN                  | Adaptive gating             | 0.9843                                   | 0.0025                      | 0.1065                    | 1.0908              | 0.9818              | 0.1041                 | 1.0883               | 127.0067              |
| Without SF-Mamba                  | AdaWAT only             | SSGCN                  | Adaptive gating             | <u>0.9865</u>                            | <b>0.0013</b>               | 0.1073                    | 1.0939              | <b>0.9852</b>       | 0.1060                 | <u>1.0926</u>        | 285.0088              |
| Without SSGCN                     | AdaWAT + SF-Mamba       | –                      | –                           | 0.9806                                   | 0.0013                      | 0.0890                    | 1.0696              | 0.9793              | 0.0878                 | 1.0684               | 213.8833              |
| Without adaptive fusion           | AdaWAT + SF-Mamba       | SSGCN                  | Concat. + $1 \times 1$ Conv | 0.9703                                   | 0.0043                      | 0.1242                    | 1.0946              | 0.9661              | 0.1200                 | 1.0903               | 107.2586              |
| Spatial–frequency track only      | AdaWAT + SF-Mamba       | –                      | –                           | 0.9796                                   | 0.0030                      | 0.1034                    | 1.0830              | 0.9766              | 0.1004                 | 1.0800               | 39.8386               |
| Spatial–spectral track only       | –                       | SSGCN                  | –                           | 0.9816                                   | 0.0015                      | 0.0693                    | 1.0509              | 0.9802              | 0.0678                 | 1.0494               | 218.0960              |
| AdaWAT-only SF track + SSGCN      | AdaWAT only             | SSGCN                  | Adaptive gating             | 0.9784                                   | 0.0043                      | 0.1007                    | 1.0792              | 0.9741              | 0.0964                 | 1.0748               | 70.1944               |
| SF-Mamba + SSGCN (without AdaWAT) | Mamba only              | SSGCN                  | Adaptive gating             | 0.9849                                   | 0.0036                      | 0.0971                    | 1.0820              | 0.9814              | 0.0935                 | 1.0784               | 39.4965               |



**Fig. 8.** Component-level design comparison of HAD-MSF on five hyperspectral datasets in terms of  $AUC(P_d, P_f)$ . (a) AdaWAT design: Fixed Haar filters versus learnable AdaWAT. (b) SSGCN design: Learnable spectral graph versus fixed spectral adjacency. (c) Fusion design: Concatenation, weighted summation, and adaptive gating. Within each subplot, only the target component is changed, while the remaining modules are kept identical to the full HAD-MSF model.

**Table 3**

Controlled comparison of AdaWAT learnability and initialization on three remote-sensing hyperspectral datasets in terms of  $AUC(P_d, P_f)$ . Only the initialization and trainability of the AdaWAT filters are changed, while all other settings remain unchanged.

| Setting          | Pavia         | Coast         | Gulfport      | Average       |
|------------------|---------------|---------------|---------------|---------------|
| Fixed Haar       | 0.9979        | 0.9843        | 0.9826        | 0.9883        |
| Learnable Random | 0.9980        | 0.9888        | 0.9876        | 0.9915        |
| Learnable Sobel  | 0.9980        | 0.9830        | 0.9763        | 0.9858        |
| Learnable Haar   | <b>0.9991</b> | <b>0.9974</b> | <b>0.9984</b> | <b>0.9983</b> |

**Table 4**

Sensitivity analysis of AdaWAT kernel size in terms of  $AUC(P_d, P_f)$  on three remote sensing datasets and two medical hyperspectral datasets. Haar-like initialization is used and the filters remain learnable.

| Kernel size  | Pavia  | Coast  | Gulfport | MHSI-1 | MHSI-2 | Avg.   |
|--------------|--------|--------|----------|--------|--------|--------|
| $3 \times 3$ | 0.9991 | 0.9974 | 0.9984   | 0.9928 | 0.9749 | 0.9925 |
| $5 \times 5$ | 0.9990 | 0.9963 | 0.9978   | 0.9919 | 0.9584 | 0.9887 |
| $7 \times 7$ | 0.9988 | 0.9955 | 0.9903   | 0.9874 | 0.9386 | 0.9821 |

To further disentangle the effects of filter learnability and initialization, Table 3 compares four controlled AdaWAT settings on the three remote-sensing datasets. Learnable Haar achieves the highest average  $AUC(P_d, P_f)$  of 0.9983, compared with 0.9883 for Fixed Haar, indicating that task-driven adaptation of the frequency filters is beneficial under the adopted Haar-like prior. It also outperforms Learnable Random (0.9915) and Learnable Sobel (0.9858) on average. These results suggest that Haar-like initialization provides a suitable structured starting point for the evaluated remote-sensing scenes, while subsequent filter adaptation is important for obtaining the final frequency representation.

We further investigate the configurability of AdaWAT by examining the sensitivity of the kernel size under Haar-like initialization with learnable filters. The results are summarized in Table 4.

As shown in Table 4, the  $3 \times 3$  setting achieves the best  $AUC(P_d, P_f)$  on all five datasets. Increasing the kernel size to  $5 \times 5$  or  $7 \times 7$  does not improve performance; instead, the average  $AUC(P_d, P_f)$  decreases from 0.9925 to 0.9887 and 0.9821, respectively. This trend is particularly evident on Gulfport, MHSI-1, and MHSI-2, indicating that larger kernels are less suitable for preserving anomaly-sensitive local frequency structures. Considering the trade-off among detection performance, computational efficiency, and model simplicity, we adopt  $3 \times 3$  learnable filters as the default setting.

We also examined the sensitivity of the SF-Mamba branch depth by varying the number of stacked Visual State-Space (VSS) blocks from 1 to 9 on the Coast dataset. The resulting  $AUC(P_d, P_f)$  remained highly stable

**Table 5**

Cross-scene evaluation on the two medical hyperspectral datasets. The model is trained on one source scene and directly evaluated on the other target scene without updating any trainable network parameters. Both scenes use the same 64-band spectral configuration, and no target-specific band reselection is performed.

| Setting                     | $AUC(P_d, P_f) \uparrow$ | $AUC(P_f, \tau) \downarrow$ | $AUC(P_d, \tau) \uparrow$ |
|-----------------------------|--------------------------|-----------------------------|---------------------------|
| MHSI-1 $\rightarrow$ MHSI-2 | 0.9485                   | 0.4088                      | 0.6826                    |
| MHSI-2 $\rightarrow$ MHSI-1 | 0.4328                   | 0.1944                      | 0.2642                    |

across different depths, indicating that HAD-MSF is relatively insensitive to this hyperparameter. Since deeper configurations did not provide a consistent performance gain while increasing computational cost, we adopt a single VSS block as the default setting.

Having validated the main component-level design choices of HAD-MSF, we next examine whether the proposed framework remains effective under a cross-scene setting.

#### 4.5. Cross-scene evaluation

To further assess the robustness of HAD-MSF beyond the in-scene setting, we conduct a cross-scene evaluation on the two medical hyperspectral datasets. The model is trained on one source scene and directly evaluated on the other target scene without updating any trainable parameters. MHSI-1 and MHSI-2 share the same 64-band spectral configuration, and no target-specific band reselection is performed. Anomaly scores are independently normalized from each target-scene residual map using unsupervised min-max scaling, without using target-scene anomaly labels.

As shown in Table 5, the transfer performance is clearly asymmetric. MHSI-1  $\rightarrow$  MHSI-2 achieves  $AUC(P_d, P_f) = 0.9485$ ,  $AUC(P_f, \tau) = 0.4088$ , and  $AUC(P_d, \tau) = 0.6826$ , whereas MHSI-2  $\rightarrow$  MHSI-1 achieves  $AUC(P_d, P_f) = 0.4328$ ,  $AUC(P_f, \tau) = 0.1944$ , and  $AUC(P_d, \tau) = 0.2642$ .

This pronounced directional asymmetry is more likely attributable to scene-level distribution differences than to spectral preprocessing, since both transfer directions use the same fixed 64-band configuration. In particular, MHSI-2 contains fewer anomalous pixels and represents a sparser anomaly scenario than MHSI-1. Differences in tissue composition, background appearance, spectral response, and local spatial structure may further contribute to the weaker MHSI-2  $\rightarrow$  MHSI-1 transfer performance.

The asymmetric transfer results indicate that cross-scene performance is strongly affected by scene-level distribution shift under the current fixed spectral configuration. This observation motivates future work on domain adaptation and distribution alignment to improve cross-scene robustness.



**Table 6**

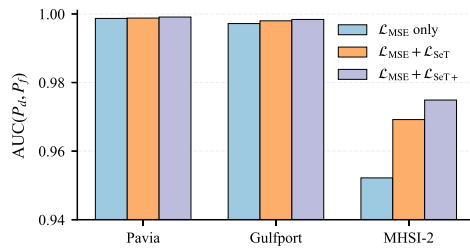
Efficiency comparison of HAD-MSF and five representative deep-learning baselines on the Gulfport dataset under band-selected (64-band) and original full-band (191-band) settings. Parameters, FLOPs, inference time per image, and peak GPU memory are reported.

| Method  | Band-selected Gulfport (64 bands) |             |                              |                        | Original Gulfport (191 bands) |             |                              |                        |
|---------|-----------------------------------|-------------|------------------------------|------------------------|-------------------------------|-------------|------------------------------|------------------------|
|         | Params (M) ↓                      | FLOPs (G) ↓ | Inference time / image (s) ↓ | Peak GPU memory (GB) ↓ | Params (M) ↓                  | FLOPs (G) ↓ | Inference time / image (s) ↓ | Peak GPU memory (GB) ↓ |
| GT-HAD  | 0.0928                            | 15.2882     | 0.003646                     | 0.3461                 | 0.2392                        | 39.5583     | 0.004598                     | 0.6153                 |
| Auto-AD | 3.0515                            | 10.3570     | 0.007453                     | 0.1675                 | 3.2304                        | 11.7388     | 0.006659                     | 0.1979                 |
| BockNet | 0.6238                            | 23.5110     | 0.003659                     | 0.2935                 | 0.7133                        | 29.6883     | 0.003433                     | 0.3641                 |
| PUNNet  | 2.4643                            | 7.1017      | 0.0084                       | 0.0508                 | 2.4888                        | 7.3456      | 0.0086                       | 0.0555                 |
| NL2Net  | 0.9693                            | 7.5366      | 0.0079                       | 0.0563                 | 0.9816                        | 7.6586      | 0.0093                       | 0.0704                 |
| HAD-MSF | 0.3731                            | 2.7354      | 0.063463                     | 0.1296                 | 2.8739                        | 23.2008     | 0.063969                     | 0.4042                 |

**Table 7**

Fixed-threshold evaluation of HAD-MSF on three representative datasets. Precision, Recall, and F1-score are reported on the held-out test pixels under two thresholding protocols: Otsu thresholding and validation-based threshold selection.

| Protocol         | Method  | Pavia     |        |        | Coast     |        |        | MHSI-2    |        |        |
|------------------|---------|-----------|--------|--------|-----------|--------|--------|-----------|--------|--------|
|                  |         | Precision | Recall | F1     | Precision | Recall | F1     | Precision | Recall | F1     |
| Otsu             | HAD-MSF | 1.0000    | 0.0526 | 0.1000 | 0.9524    | 0.3704 | 0.5333 | 0.2222    | 0.0110 | 0.0210 |
| Validation-based | HAD-MSF | 0.9545    | 0.3684 | 0.5316 | 0.8519    | 0.4259 | 0.5679 | 0.4095    | 0.4490 | 0.4284 |



**Fig. 9.** Validation of the enhanced separation training strategy on representative datasets in terms of  $AUC(P_d, P_f)$ . All variants use the same network backbone, and only the training objective is changed.

#### 4.6. Validation of enhanced separation training

To further evaluate the effect of the proposed soft anomaly mask, Fig. 9 compares three training objectives: the reconstruction loss alone, the reconstruction loss combined with the original separation training based on a hard binary mask, and the reconstruction loss combined with the proposed softened separation training. For fair comparison, all variants use the same network backbone, and only the loss formulation is changed.

As shown in Fig. 9, introducing separation training consistently improves the results over using the reconstruction loss alone. More importantly,  $\mathcal{L}_{Set+}$  further outperforms the original  $\mathcal{L}_{Set}$  on all representative datasets in terms of  $AUC(P_d, P_f)$ . These results support the use of a continuous soft anomaly mask, which leads to a more stable optimization process and improves the practical quality of residual-based anomaly scoring.

Since the proposed loss formulation is intended to improve the stability of residual-based anomaly scoring, we next examine its practical behavior under explicit thresholding.

#### 4.7. Fixed-threshold evaluation and threshold selection

In the main paper, method comparison is primarily based on threshold-free AUC metrics, which avoid bias caused by a single manually selected operating point. To further address the practical issue of threshold selection, we additionally evaluate fixed-threshold performance under two protocols: Otsu thresholding and validation-based threshold selection. For fair comparison, each anomaly score map is first normalized to the range [0, 1].

The validation-based threshold is used only for post-hoc operating-point calibration and does not participate in model training or the primary ROC/AUC evaluation. Without labeled validation pixels, HAD-MSF can directly output continuous anomaly scores or use an unsupervised thresholding strategy such as Otsu thresholding.

For validation-based threshold selection, the pixels of each dataset are randomly partitioned into a validation subset (20%) and a held-out test subset (80%) using stratified sampling over the anomaly and background labels with a fixed random seed of 42. The decision threshold is selected from 401 quantile-based candidates by maximizing the pixel-level F1-score using only the validation pixels and is then fixed for evaluation on the held-out test pixels. For Otsu thresholding, the threshold is determined from the complete normalized anomaly score map without using anomaly/background labels, and its performance is evaluated on the same held-out test pixels. The label-based validation split is used only for threshold calibration and evaluation and is not involved in the unsupervised training of HAD-MSF.

Although the reconstruction error is used as both the training signal and the test-time anomaly score, the model is not trained to fit anomaly labels directly. Instead, HAD-MSF is optimized to reconstruct the background while suppressing anomaly responses through the enhanced separation training objective. Therefore, the residual map serves as a test-time discrepancy measure rather than a label-dependent prediction target.

As shown in Table 7, validation-based threshold calibration improves the held-out test F1-score from 0.1000 to 0.5316 on Pavia, from 0.5333 to 0.5679 on Coast, and from 0.0210 to 0.4284 on MHSI-2 compared with Otsu thresholding. The improvement is particularly evident on Pavia and MHSI-2, where Otsu thresholding yields high precision but very low recall. These results indicate that calibration on a separate validation subset can provide a more appropriate operating point for highly imbalanced anomaly distributions, while the reported performance remains evaluated exclusively on held-out test pixels.

It should be noted that these fixed-threshold Precision/Recall/F1 results are not directly comparable in scale to the threshold-free AUC metrics reported in the main experiments. While the AUC-based measures evaluate ranking quality across all possible thresholds, the fixed-threshold metrics are point estimates after hard thresholding and are therefore more sensitive to class imbalance and threshold selection. Accordingly, the fixed-threshold results are reported as a complementary operating-point analysis rather than as the primary evaluation protocol.

**Table 8**

Comparison of representative deep-learning-based HAD methods under band-selected (64-band) and original full-band settings on three remote sensing datasets. The best results are shown in bold, and the second-best results are underlined within each setting.

| Method  | Pavia         |               | Coast         |               | Gulfport      |               | Average       |               |
|---------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|         | 64            | Full          | 64            | Full          | 64            | Full          | 64            | Full          |
| GT-HAD  | <b>0.9992</b> | 0.9993        | 0.9968        | <b>0.9974</b> | 0.9931        | 0.9959        | 0.9964        | 0.9975        |
| Auto-AD | 0.9900        | 0.9912        | 0.9914        | 0.9865        | 0.9738        | 0.9659        | 0.9851        | 0.9812        |
| BockNet | 0.9988        | 0.9942        | 0.9917        | 0.9860        | <u>0.9933</u> | 0.9303        | 0.9946        | 0.9702        |
| PUNNet  | 0.9982        | 0.9938        | 0.9863        | 0.9792        | 0.9599        | 0.9755        | 0.9815        | 0.9828        |
| NL2Net  | 0.9918        | 0.9965        | 0.9798        | 0.9954        | 0.9877        | 0.9861        | 0.9864        | 0.9927        |
| HAD-MSF | <u>0.9991</u> | <b>0.9994</b> | <b>0.9974</b> | <u>0.9972</u> | <b>0.9984</b> | <b>0.9985</b> | <b>0.9983</b> | <b>0.9984</b> |

**Table 9**

Noise-robustness evaluation of HAD-MSF on the coast and Gulfport datasets under Gaussian and stripe noise. Three representative AUC-based metrics are reported as averages over five independent runs.

| Noise Level              | Coast                       |                                |                              | Gulfport                    |                                |                              |
|--------------------------|-----------------------------|--------------------------------|------------------------------|-----------------------------|--------------------------------|------------------------------|
|                          | $AUC_{(P_d, P_f)} \uparrow$ | $AUC_{(P_f, \tau)} \downarrow$ | $AUC_{(P_d, \tau)} \uparrow$ | $AUC_{(P_d, P_f)} \uparrow$ | $AUC_{(P_f, \tau)} \downarrow$ | $AUC_{(P_d, \tau)} \uparrow$ |
| Gaussian $\sigma = 0.01$ | 0.9968                      | 0.00042                        | 0.2121                       | 0.9777                      | 0.0010                         | 0.0477                       |
| Gaussian $\sigma = 0.05$ | 0.9963                      | 0.00036                        | 0.1947                       | 0.9772                      | 0.0010                         | 0.0498                       |
| Gaussian $\sigma = 0.10$ | 0.9963                      | 0.00036                        | 0.1944                       | 0.9770                      | 0.0010                         | 0.0558                       |
| Gaussian $\sigma = 0.20$ | 0.9964                      | 0.00035                        | 0.1938                       | 0.9758                      | 0.0011                         | 0.0501                       |
| Gaussian $\sigma = 0.30$ | 0.9963                      | 0.00037                        | 0.1956                       | 0.9772                      | 0.0011                         | 0.0609                       |
| Stripe Mild (5%)         | 0.9958                      | 0.00034                        | 0.1819                       | 0.9764                      | 0.0011                         | 0.0623                       |
| Stripe Moderate (10%)    | 0.9956                      | 0.00033                        | 0.1624                       | 0.9604                      | 0.0010                         | 0.0011                       |
| Stripe Severe (20%)      | 0.9962                      | 0.00035                        | 0.1917                       | 0.9688                      | 0.0009                         | 0.0314                       |

#### 4.8. Efficiency analysis

To provide a controlled comparison of computational efficiency, Table 6 reports the number of parameters, FLOPs, per-image inference time, and peak GPU memory for HAD-MSF and five representative deep-learning baselines on the Gulfport dataset under the band-selected (64-band) and original full-band (191-band) settings.

We further examine whether SIFDR-based band selection affects detection performance. As shown in Table 8, the effect of band selection varies across methods and datasets. For HAD-MSF, however, the absolute difference in  $AUC(P_d, P_f)$  between the 64-band and full-band settings is only 0.0003 on Pavia, 0.0002 on Coast, and 0.0001 on Gulfport. Under the full-band setting, HAD-MSF still achieves the highest average  $AUC(P_d, P_f)$  of 0.9984. These results indicate that SIFDR mainly serves as an efficiency-oriented preprocessing strategy rather than a prerequisite for HAD-MSF.

Although HAD-MSF achieves the lowest FLOPs among the compared methods, this advantage does not directly translate into the shortest inference time. Under the 64-band setting, HAD-MSF requires 0.0635 s per image, compared with 0.0084 s for PUNNet and 0.0079 s for NL2Net, and its peak GPU memory usage is also not the lowest. Therefore, the computational advantage of HAD-MSF should be interpreted primarily in terms of arithmetic complexity and model compactness rather than runtime latency. Future work will investigate deployment-oriented optimization, including pruning, quantization, and GPU-specific acceleration.

#### 4.9. Robustness analysis

To further assess robustness under more diverse hyperspectral degradations, Table 9 evaluates HAD-MSF on both Coast and Gulfport under Gaussian and structured stripe noise. Three representative AUC metrics are reported, with noisy-condition results averaged over five independent runs. Gaussian perturbations follow the original noise protocol, while the stripe settings corrupt 5%, 10%, and 20% of image columns for the mild, moderate, and severe cases, respectively. Noise is introduced before band selection.

HAD-MSF remains relatively stable under the tested perturbations. On Coast,  $AUC_{(P_d, P_f)}$  stays within 0.9963–0.9968 under Gaussian noise and 0.9956–0.9962 under stripe noise. On Gulfport, the corresponding ranges are 0.9758–0.9777 and 0.9604–0.9764, respectively. The  $AUC_{(P_f, \tau)}$  values also remain low across the tested conditions, indicating generally stable false-alarm control. Overall, these experiments extend the original single-dataset Gaussian-noise analysis to an additional dataset and a representative structured degradation, while also showing that robustness is scene dependent.

## 5. Conclusion

In this work, we propose HAD-MSF, a neural reconstruction framework for hyperspectral anomaly detection that jointly models spatial, spectral, and frequency information. The methodological contribution of HAD-MSF lies in a problem-specific structural factorization in which frequency decomposition captures smooth and localized responses, state-space modeling provides long-range context, graph propagation models spatial-spectral topology, and adaptive gating integrates these complementary representations within a label-free reconstruction objective. HAD-MSF combines AdaWAT and SF-Mamba for frequency-aware long-range dependency modeling with SSGCN for spatial-spectral representation, and adaptively integrates the complementary features through a lightweight gating mechanism. Experiments on three remote sensing and two medical hyperspectral datasets demonstrate consistently competitive detection performance while maintaining relatively low arithmetic complexity. Additional analyses of component design, enhanced separation training, threshold selection, cross-scene transfer, and full-band applicability further validate the effectiveness and robustness of the proposed framework.

Several limitations remain. The medical evaluation is currently limited to two intraoperative scenes, and the asymmetric cross-scene results indicate that distribution shift remains challenging. In addition, the current AdaWAT employs a single-level  $3 \times 3$  decomposition, while SSGCN uses fixed local spectral adjacency, which may limit adaptability to sensors with different spectral characteristics. Finally, although HAD-MSF achieves low arithmetic complexity, its measured inference latency is

not the lowest among the compared deep baselines; its computational advantage should therefore be interpreted as an accuracy-complexity trade-off rather than universal runtime superiority. Future work will investigate cross-dataset adaptation, broader full-band validation, and hardware-oriented optimization to further improve the generalizability and practical deployment of HAD-MSF.

## CRediT authorship contribution statement

**Jinzhuang Xu:** Writing – original draft, Visualization, Methodology, Conceptualization; **Chenglong Zhang:** Writing – review & editing, Validation, Supervision, Project administration; **Xiaoxue Wang:** Writing – review & editing, Visualization, Supervision, Resources; **Xiaoli Yang:** Writing – review & editing, Visualization, Conceptualization; **Yuping Han:** Writing – review & editing, Methodology, Formal analysis; **Xi-aopeng Ma:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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